

Mohamed Khalil ZRELLY

Generative AI Engineer Apprentice | Computer Science Engineering Student

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Professional Summary

Generative AI Engineer Apprentice at Crédit Agricole, specializing in agentic workflow orchestration. Computer Science Engineering student at Sup Galilée, Sorbonne Paris North University, with hands-on expertise in RAG, NLP, and reinforcement learning.

Work Experience

Crédit Agricole Personal Finance & Mobility

France

Generative AI Engineer Apprentice

September 2025 — Present · 10 months

- Developed a declarative, config-driven microservice for **workflow management and orchestration**, making it easy to manage and execute complex AI workflows at scale.
- Built an automated **workflow evaluation engine** leveraging **LLM-as-a-judge** to assess outputs against custom metrics, with a **dashboard** to analyse workflow behaviour.

Agentic Workflows · Advanced RAG · Amazon Bedrock · LLM Observability · Python/FastAPI · Amazon DynamoDB · Python/Streamlit

VERMEG for Banking & Insurance Software

Tunisia

Natural Language Processing Intern

February 2024 — June 2024 · 5 months

- Implemented a system that automates email classification and information extraction from attachments, using fine-tuned **BERT** and **OCR** to improve accuracy and optimise processing.
- Developed a web application for **BERT Fine-Tuning**, email management, and real-time data analysis and visualisation, enabling model management and performance monitoring.

Hugging Face Transformers · Sentence-Transformers · Fine-Tuning · PyTorch · Tesseract OCR · Python/Flask · MongoDB · React/Redux

Sofiatech

Tunisia

Machine Learning Intern

June 2023 — August 2023 · 3 months

- Evaluated multiple **reinforcement learning** algorithms from **Stable Baselines 3** across various **Gymnasium** environments to assess agent performance and training efficiency.
- Built a **2D game** in **Pygame** where a **Deep Q-Learning** agent learns to navigate a dynamic environment, applying the best-performing algorithm from the benchmarking phase.

Reinforcement Learning · Deep Q-Learning · stable-baselines3 · TensorFlow/Keras · Gymnasium · NumPy · Pandas · Matplotlib · Python/Pygame

Featured Projects

CVision | RAG · CV Recommendation System

June 2025 — August 2025

Created an intelligent **CV recommendation system** based on job description using **Retrieval-Augmented Generation (RAG)**, to optimise the selection and automatic ranking of the most relevant profiles.

Advanced RAG · Re-Ranking · LangChain · Qwen3-VL · Milvus · Python/Streamlit

Twixel | Multi-Agent · Social Listening System

In Development — 2026

Designing an intelligent **multi-agent system** for automated social listening on X (Twitter), aimed at identifying emerging trends in real time and generating actionable marketing campaign ideas at scale.

Multi-Agent Systems · LangGraph · MCP/A2A · GPT-OSS · Qdrant · Python/FastAPI · Vue.js

Technical Skills

Machine Learning, DL & MLOps	PyTorch, TensorFlow, Scikit-Learn, XGBoost, Amazon SageMaker
Generative AI & Agentic Systems	LangChain, LangGraph, Milvus, Qdrant, Amazon Bedrock
Data Analysis & Engineering	Pandas, Matplotlib, PySpark, Apache Airflow, Amazon S3
SQL & NoSQL Databases	Oracle, MySQL, PostgreSQL, MongoDB, Amazon DynamoDB
Full-Stack Web Development	FastAPI, Flask, Express.js, Laravel, React, Vue.js
Version Control, Testing & QA	Git, GitHub, Pytest, Jest, Vitest, Playwright, Postman
Cloud Infrastructure & DevOps	AWS, Docker, Kubernetes, Terraform, GitHub Actions
Programming Languages	Python, JavaScript/TS, PHP, Java, C/C++, PL/SQL

Education

Sup Galilée – Sorbonne Paris North University

Villetaneuse, France

Engineering Degree in Computer Science

2024 — 2027 (Expected)

ISI Ariana – University of Tunis El Manar

Ariana, Tunisia

Bachelor of Science in Computer Science

2021 — 2024 · High Distinction

Certifications & Languages

Microsoft Azure	Azure AI Fundamentals (AI-900) — Jan 2024 · Azure Data Fundamentals (DP-900) — Jan 2024
Udemy	Spark and Python for Big Data with PySpark — Mar 2024 · NLP with Python — Feb 2024
Spoken Languages	Arabic (Native) · French B2 — TCF · English B2 — TOEIC